Quiz-1

1- A and B are two independent events such that $P(\bar{A}) = 0.7$, $P(\bar{B}) = K$ and $P(A \cup B) = 0.8$, then K is

(a) 5/7 (b) 1, (c) 2/7 (d) NOT

2- If $P(A \cap B) = 1/2$, $P(\bar{A} \cap \bar{B}) = 1/3$, and $P(A) = P(B) = p$, then the value of p is given by

(a) 1/2 (b) 7/8 (c) 1/3 (d) 7/12

3. If A and B are mutually exclusive events, then

(a) $P(A \cup B) = P(A) \cdot P(B)$ (b) $P(A \cup B) = P(A) + P(B)$

(c) $P(A \cup B) = 0$.

(4) The mean of the binomial distribution ${}^{10}C_x \left(\frac{2}{3}\right)^x \left(\frac{3}{5}\right)^{10-x}$; $x=0,1,2,3,\ldots 10$ is

(a) 4 (b) 6 (c) 5 (d) 0

(5) If $X \sim$ Poisson(λ), then $P(X=3) = P(X=2)$. implies

(a) $\lambda=2$, (b) $=3$ (c) $=4$, (d) NOT

(6) If $F(x) = \begin{cases} 0, & x<0 \\ 0.3 & 0 \le x < 2 \\ 0.8 & 2 \le x < 5 \\ 1 & x \ge 5 \end{cases}$

then $P(X=2)$

(a) 0.3, (b) 0.5 (c) $=0.8$ (d) $=0.2$

(7) If $M_X(t) = (0.6 + 0.4e^t)^8$, then X has

A. Binomial distribution with $n=8$, $p=0.4$

B. Poisson "

C. Bernoulli "

D. Normal "